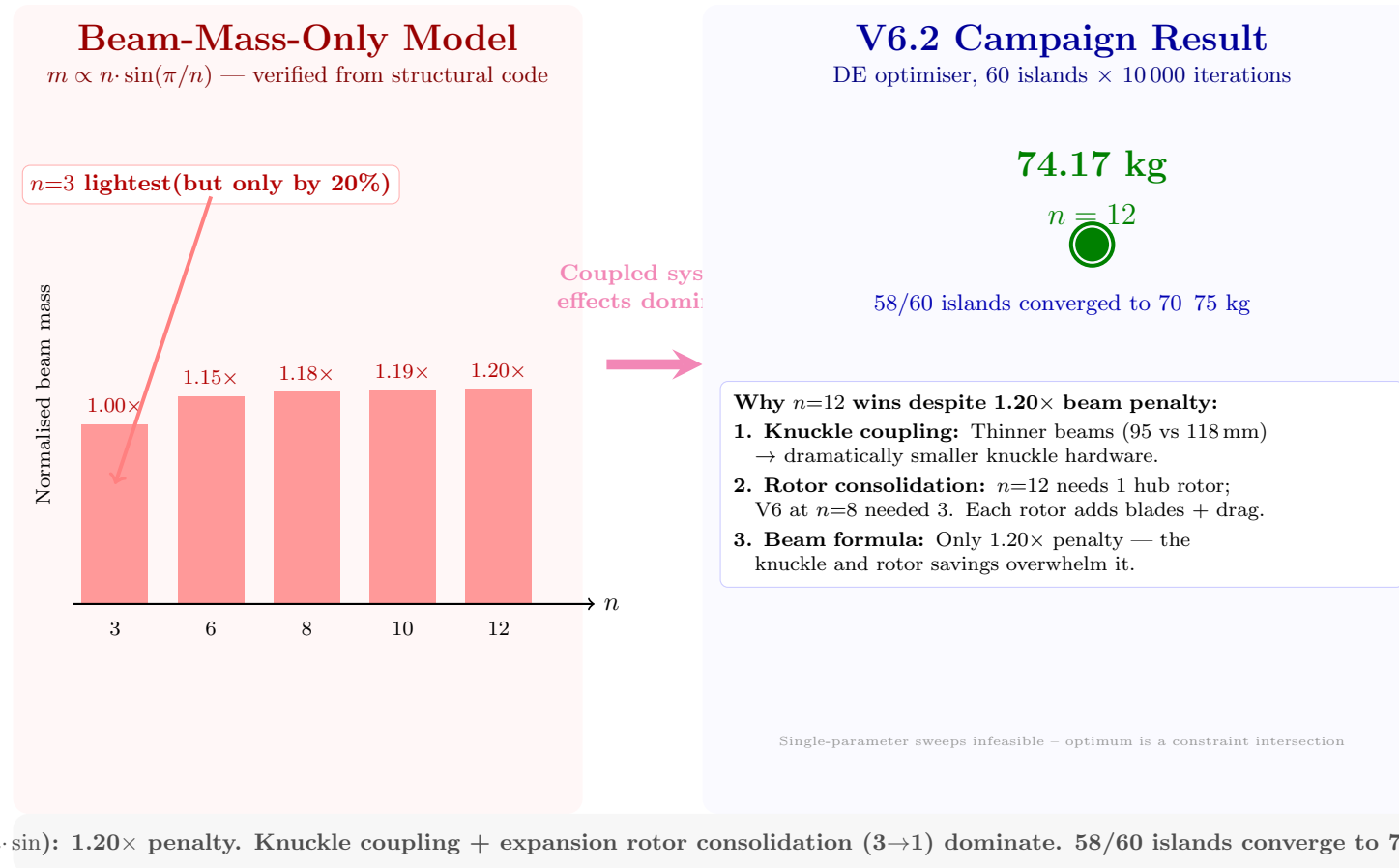


Why $n=12$ Wins: Beam Formula vs System Reality

Beam formula ($n \cdot \sin(\pi/n)$): only $1.20\times$ penalty. Knuckle coupling + expansion rotor consolidation dominate.



Beam-only ($n \cdot \sin$): $1.20\times$ penalty. Knuckle coupling + expansion rotor consolidation ($3 \rightarrow 1$) dominate. 58/60 islands converge to 74.17 kg at $n=12$.